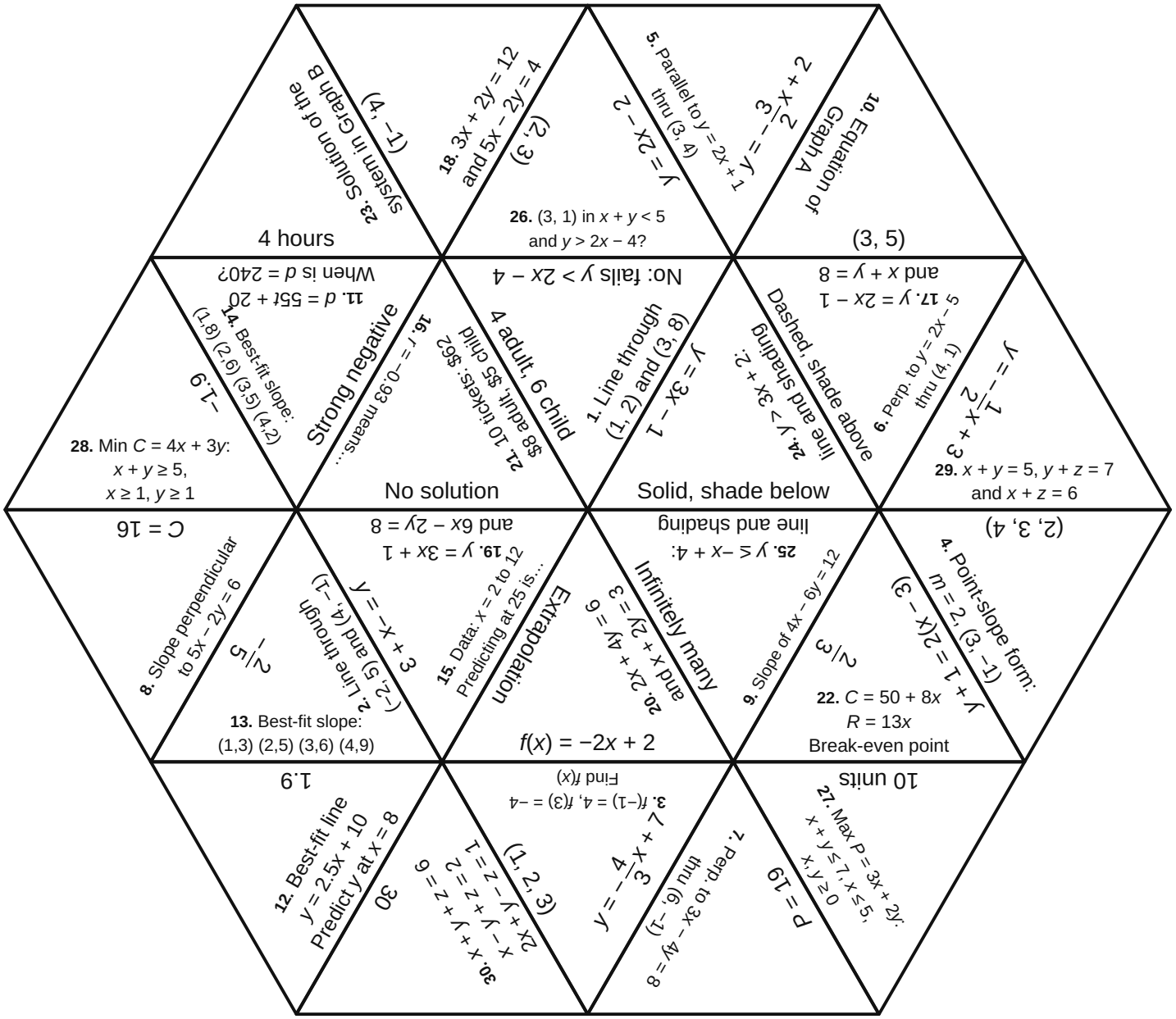
Linear Functions & Systems Tarsia: Answer Key

Solved hexagon. Each shared edge pairs a problem with its answer.



Answer List

Linear Functions & Systems Tarsia · all 30 matches

#	Problem	Answer (on the matching edge)
1	Line through (1, 2) and (3, 8)	$y = 3x - 1$
2	Line through (−2, 5) and (4, −1)	$y = -x + 3$
3	$f(-1) = 4$, $f(3) = -4$ Find $f(x)$	$f(x) = -2x + 2$
4	Point-slope form: $m = 2$, (3, −1)	$y + 1 = 2(x - 3)$
5	Parallel to $y = 2x + 1$ thru (3, 4)	$y = 2x - 2$
6	Perp. to $y = 2x - 5$ thru (4, 1)	$y = -\frac{1}{2}x + 3$
7	Perp. to $3x - 4y = 8$ thru (6, −1)	$y = -\frac{4}{3}x + 7$
8	Slope perpendicular to $5x - 2y = 6$	$-\frac{2}{5}$
9	Slope of $4x - 6y = 12$	$\frac{2}{3}$
10	Equation of Graph A	$y = -\frac{3}{2}x + 2$
11	$d = 55t + 20$ When is $d = 240$?	4 hours
12	Best-fit line $y = 2.5x + 10$ Predict y at $x = 8$	30
13	Best-fit slope: (1,3) (2,5) (3,6) (4,9)	1.9
14	Best-fit slope: (1,8) (2,6) (3,5) (4,2)	−1.9
15	Data: $x = 2$ to 12 Predicting at 25 is...	Extrapolation
16	$r = -0.93$ means...	Strong negative
17	$y = 2x - 1$ and $x + y = 8$	(3, 5)
18	$3x + 2y = 12$ and $5x - 2y = 4$	(2, 3)
19	$y = 3x + 1$ and $6x - 2y = 8$	No solution
20	$2x + 4y = 6$ and $x + 2y = 3$	Infinitely many
21	10 tickets: \$62 \$8 adult, \$5 child	4 adult, 6 child
22	$C = 50 + 8x$ $R = 13x$ Break-even point	10 units

23	Solution of the system in Graph B	$(4, -1)$
24	$y > 3x + 2$: line and shading	Dashed, shade above
25	$y \le -x + 4$: line and shading	Solid, shade below
26	$(3, 1)$ in $x + y < 5$ and $y > 2x - 4$?	No: fails $y > 2x - 4$
27	Max $P = 3x + 2y$: $x + y \le 7$, $x \le 5$, $x, y \ge 0$	$P = 19$
28	Min $C = 4x + 3y$: $x + y \ge 5$, $x \ge 1$, $y \ge 1$	$C = 16$
29	$x + y = 5$, $y + z = 7$ and $x + z = 6$	$(2, 3, 4)$
30	$x + y + z = 6$ $x - y + z = 2$ $2x + y - z = 1$	$(1, 2, 3)$